

Critical Evaluation of Phishing URL Detection

A leakage-aware, reproducible machine-learning benchmark

PROJECT	Data Science in Cybersecurity
SUBMISSION	raghdans/phishing_url_detection_submission2
EVALUATED SOURCE	vaibhavbichave/Phishing-URL-Detection
REPRODUCIBILITY SEED	42
REPORT DATE	4 July 2026

Central finding

Random Forest remains the strongest tested model after strict leakage control, achieving F1 0.9338 and ROC-AUC 0.9889 on an untouched, group-separated final test set. The lower result relative to the earlier random-split estimate is evidence of a more credible evaluation, not a weaker project.

Prepared as a reproducible academic evaluation. Results describe this dataset and do not establish production readiness.

Executive Summary

This study reproduces and extends a public phishing URL classification project. It compares Logistic Regression, Random Forest, and Gradient Boosting using 30 pre-engineered numerical indicators. Phishing is encoded as the positive class so recall and false negatives have direct security meaning.

The main methodological contribution is control of exact-feature duplication. The 11,054-row dataset contains only 5,785 unique predictor vectors; 5,269 rows repeat a vector seen elsewhere, and 64 vector groups contain conflicting labels. A conventional random split can place identical vectors in training and testing, creating information leakage. This evaluation assigns every identical vector to one group and keeps each group within a single partition and cross-validation fold.

Model selection uses five-fold stratified group cross-validation on development data. The Random Forest is selected by mean phishing-class F1. Its decision threshold is then selected from out-of-fold development probabilities, maximizing phishing recall while requiring at least 95% phishing precision. Only after those choices are fixed is the untouched final test partition evaluated.

Final model	Threshold	Accuracy	Precision	Recall	F1	MCC	ROC-AUC
Random Forest	0.55	0.9416	0.9493	0.9188	0.9338	0.8820	0.9889

The final test contains 81 missed phishing samples and 49 blocked legitimate samples. Strong ranking performance does not remove the operational cost of either error type.

1. Research Question and Scope

The research question is: How reliably do a linear baseline and nonlinear ensembles distinguish phishing from legitimate websites on the supplied feature dataset when model selection, threshold selection, and duplicate leakage are explicitly controlled?

The scope is deliberately narrower than real-time URL detection. The CSV contains already-computed indicators rather than raw URLs. Consequently, this study evaluates classification under the dataset's representation; it does not validate feature collection latency, live domain lookup reliability, or resistance to adaptive attackers.

2. Source and Dataset Evaluation

2.1 Selected source

The selected source is the public GitHub repository Phishing-URL-Detection by Vaibhav Bichave. It provides code and a CSV, making basic reproduction possible. Its documentation does not, however, fully establish URL collection dates, sampling frames, label adjudication, or whether related observations were grouped during evaluation. Those omissions limit external validity and make high headline accuracy insufficient evidence for deployment.

2.2 Data-quality audit

Audit item	Result	Interpretation
Rows	11,054	Sample-level observations
Model predictors	30	Identifier and target excluded
Missing values	0	No imputation activated on this snapshot
Unique feature vectors	5,785	Substantial exact repetition
Rows beyond first group member	5,269	Potential random-split leakage
Conflicting-label groups	64	Identical predictors, inconsistent targets

The identifier column previously masked the duplication because every row identifier is unique. Auditing duplicates only across all raw columns therefore returns zero and is misleading for modeling. The relevant audit is performed after removing the identifier and, separately, after removing both identifier and target.

Conflicting-label groups imply irreducible ambiguity under the available representation: the model receives the same predictors for samples assigned different labels. This may reflect label noise, coarse feature discretization, or unobserved information. The study retains these rows but prevents their groups from crossing folds.

2.3 Label semantics

The source labels phishing as -1 and legitimate as 1. This reproduction maps phishing to 1 and legitimate to 0. Therefore phishing recall measures the proportion of phishing samples detected, and a false negative is a phishing sample incorrectly allowed as legitimate. This convention removes the ambiguity present in the earlier report.

3. Experimental Design

3.1 Leakage-aware partitions

Exact predictor vectors are factorized into groups. StratifiedGroupKFold with five folds, shuffle enabled, and random state 42 supplies candidate outer partitions. The candidate closest to the desired 20% size and global phishing rate becomes the final test partition. This yields 8,827 development rows and 2,227 final-test rows, with zero shared groups.

Partition	Rows	Phishing rate	Used for
Development	8,827	0.4417	CV model selection and OOF threshold selection
Final test	2,227	0.4481	One evaluation after all choices are fixed

3.2 Models and preprocessing

Logistic Regression provides an interpretable linear baseline and uses median imputation plus standardization inside its pipeline. Random Forest uses 300 trees, class balancing, and a fixed seed. Gradient Boosting supplies a second nonlinear comparator. Tree pipelines include median imputation but no scaling. Fitting preprocessing inside each fold prevents validation information from affecting training transformations.

3.3 Metrics and uncertainty

Accuracy is reported for comparability, but the primary selection metric is phishing-class F1. Precision measures how often phishing alerts are correct; recall measures how much phishing is detected. MCC uses all four confusion-matrix cells and is informative under class imbalance. ROC-AUC evaluates ranking across thresholds. Fold means, sample standard deviations, and normal-approximation 95% intervals summarize split variability. With only five correlated training folds, these intervals are descriptive rather than population guarantees.

4. Development Results

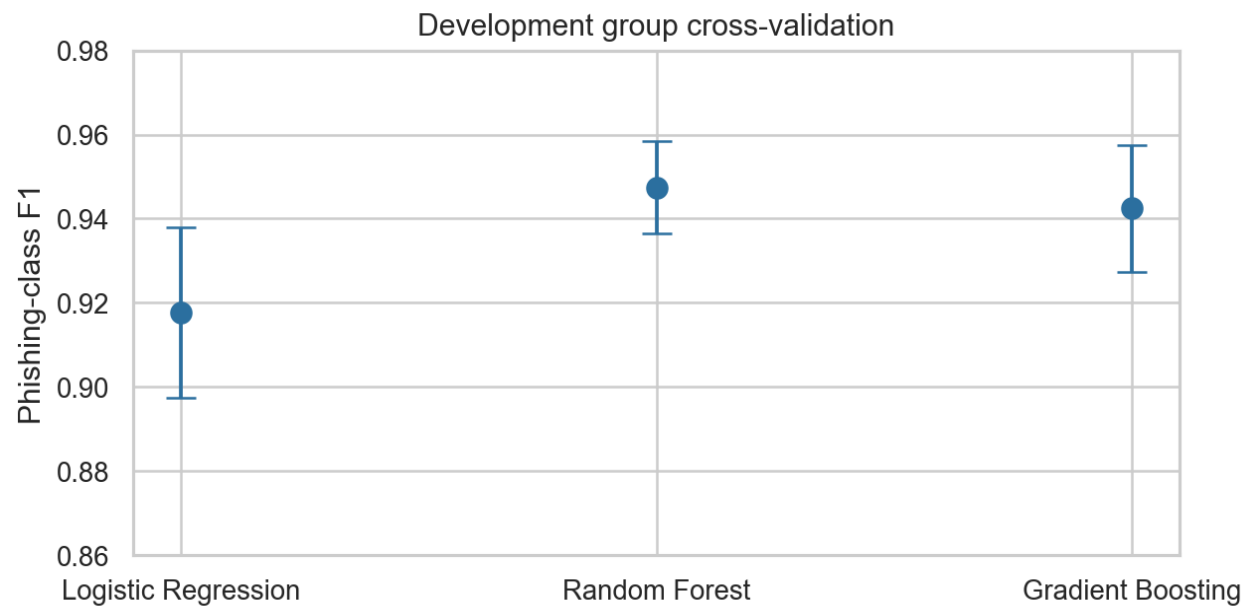


Figure 1. Mean phishing-class F1 with approximate 95% intervals across grouped folds.

Model	Accuracy	Precision	Recall	F1	MCC	ROC-AUC
Random Forest	0.9538	0.9500	0.9452	0.9474	0.9064	0.9926
Gradient Boosting	0.9498	0.9518	0.9334	0.9424	0.8982	0.9899
Logistic Regression	0.9285	0.9276	0.9085	0.9177	0.8549	0.9795

Random Forest has the highest mean F1 (0.9474), followed by Gradient Boosting (0.9424) and Logistic Regression (0.9177). The intervals overlap between the two tree ensembles, so the ranking should not be overstated as universal superiority. The selected model is simply the best under the predeclared development criterion.

5. Threshold Selection

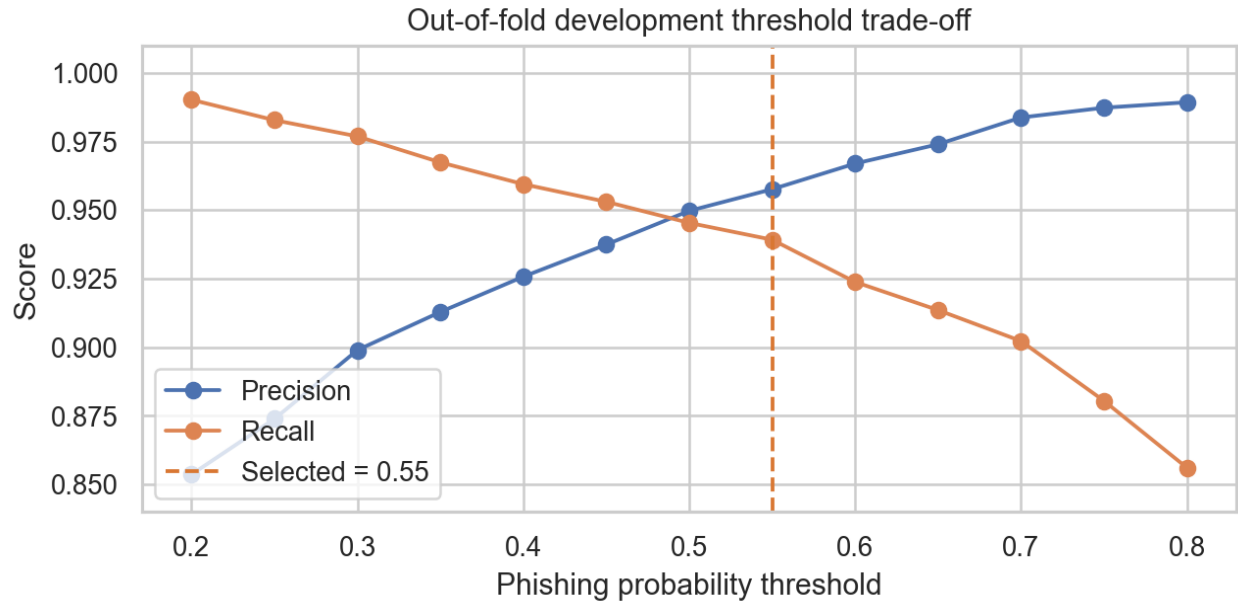


Figure 2. Threshold choice uses out-of-fold development predictions only.

The selected threshold is 0.55. On out-of-fold development predictions it produces precision 0.9576, recall 0.9392, and F1 0.9483. It is the threshold with maximum recall among candidates meeting the 0.95 precision floor. The precision floor is a transparent study assumption, not a claim about every organization's costs.

6. Final Test Evaluation

Metric	Value	Security interpretation
Accuracy	0.9416	Overall proportion correct
Phishing precision	0.9493	Reliability of phishing alerts
Phishing recall	0.9188	Fraction of phishing detected
Phishing F1	0.9338	Balance of phishing precision and recall
MCC	0.8820	Balanced correlation across all outcomes
ROC-AUC	0.9889	Threshold-independent ranking quality

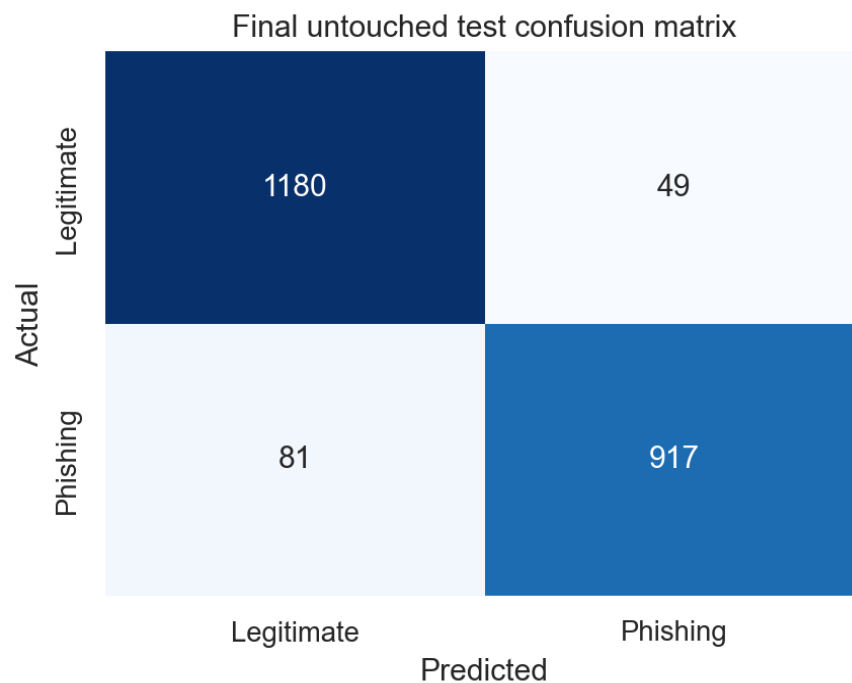


Figure 3. Phishing is the positive class; the lower-left cell contains missed phishing.

Of 998 phishing samples, 917 are detected and 81 are missed. Of 1,229 legitimate samples, 1180 are allowed and 49 are blocked. The false negatives are the higher-risk security failures because they may expose users to credential theft or malware; false positives impose usability, support, and business-continuity costs.

6.1 Why the corrected score is lower

The earlier submission reported final-test F1 near 0.971 under an ordinary random split. The leakage-aware evaluation reports 0.934. Because nearly half the rows repeat a feature vector, random splitting can let a model encounter the same vector during training and testing. Group separation removes this shortcut. The corrected result is therefore the appropriate headline figure for this submission.

7. Interpretation

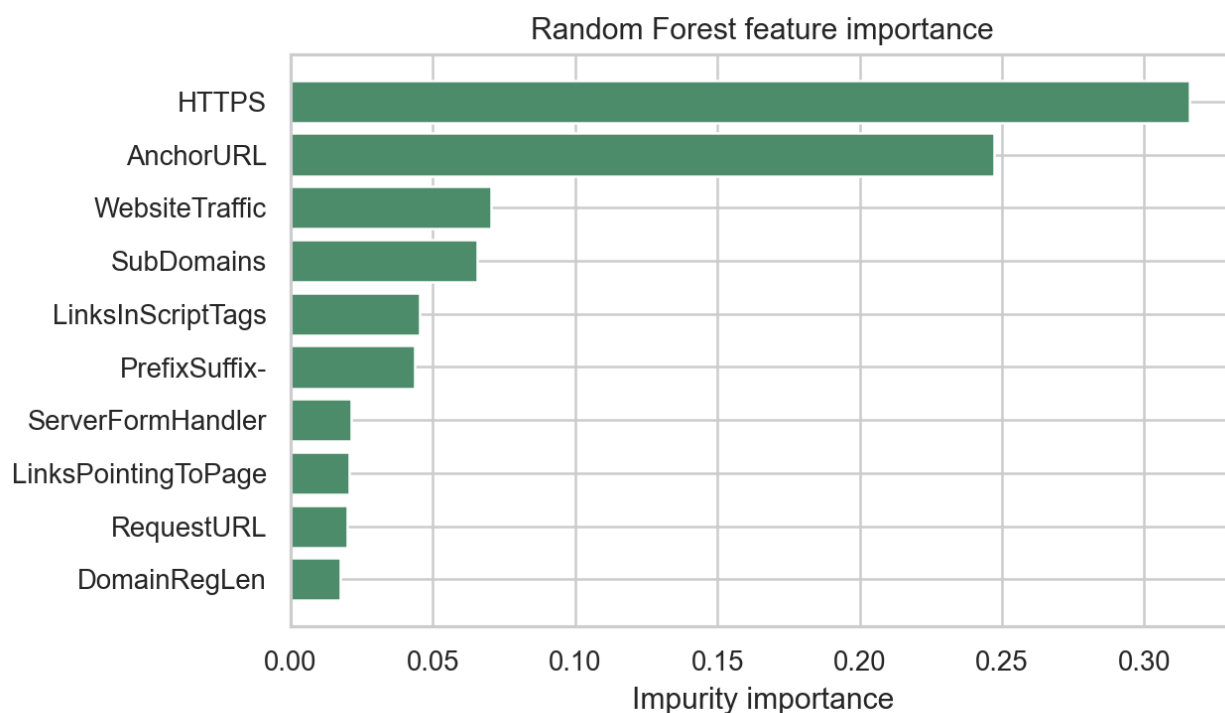


Figure 4. Impurity-based importance describes model usage, not causal effects.

HTTPS and AnchorURL dominate Random Forest importance, followed by WebsiteTraffic and SubDomains. This is consistent with the dataset's engineered representation, but it does not prove that changing one indicator causes a site to become phishing or legitimate. Impurity importance may also favor variables offering more candidate splits and can divide credit among correlated predictors. Permutation importance on future, grouped data would provide a useful robustness check.

8. Threats to Validity

Internal validity

Exact-match grouping addresses a major leakage route but cannot detect near-duplicates, shared domains, repeated campaigns, or preprocessing performed before the dataset was published. Conflicting labels also cap achievable consistency. Hyperparameters are largely fixed rather than searched, which limits optimization but reduces selection overfitting.

External validity

No timestamps support train-past/test-future evaluation. Phishing tactics, hosting, browser behavior, and reputation services drift. Random grouped folds estimate performance within this historical dataset, not on future campaigns or different organizations.

Construct and operational validity

The dataset measures discretized indicators rather than raw URL text or a complete browsing context. Some features may require live network or third-party reputation queries. Their availability, latency, failure modes, privacy implications, and susceptibility to attacker manipulation are outside the experiment.

Statistical conclusion validity

Five folds provide a limited view of variability, and fold estimates share training data. The normal-approximation intervals are therefore descriptive. A stronger study would use repeated grouped splits or a group-level bootstrap.

9. Reproducibility

The repository now matches its documentation. The dataset is under `data/`, the executed notebook is under `notebooks/`, reusable logic is under `src/`, report artifacts are under `reports/`, and tests are under `tests/`. Dependencies are version-pinned. Paths are resolved from the repository rather than from `/content` or a personal directory. Randomness is fixed, schema and label assumptions are validated, and four automated tests cover loading, label semantics, deterministic splitting, and group separation.

A reviewer can create a virtual environment, install `requirements.txt`, execute the notebook with `nbconvert`, run `pytest`, and regenerate this report with `python -m src.generate_report`. The committed notebook is already executed, while the CSV tables provide compact, machine-readable evidence for every headline result.

10. Conclusions

Random Forest is the strongest of the three tested models under grouped development cross-validation and achieves strong discrimination on the untouched final test. More important than the model ranking, however, is the correction of evaluation leakage. The project demonstrates that repository organization, label semantics, data-quality auditing, and partition design can materially change the credibility of a cybersecurity result.

The study supports a careful claim: engineered website indicators can classify this dataset with high performance when exact duplicate vectors are separated. It does not support a claim that the model is deployment-ready. Production evidence would require fresh temporally separated URLs, domain- or campaign-level grouping, calibrated probabilities, cost-based thresholds, adversarial evaluation, and continuous drift monitoring.

11. Future Work

Priority extensions are: (1) obtain timestamped raw URLs with documented sampling and label adjudication; (2) group by registered domain and campaign rather than only exact vectors; (3) quantify near-duplicate similarity; (4) evaluate calibration and decision curves; (5) use nested grouped validation for any hyperparameter search; (6) test on an external, newer dataset; and (7) measure feature extraction latency, failure, privacy, and drift.

References

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- [2] D. M. W. Powers, Evaluation: From Precision, Recall and F-Measure to ROC, Informedness, Markedness and Correlation, *Journal of Machine Learning Technologies*, 2011.
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- [4] F. Pedregosa et al., Scikit-learn: Machine Learning in Python, *Journal of Machine Learning Research*, 2011.